

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		bonds broken $(\text{C}=\text{C}) + 4(\text{C}-\text{H}) + 3(\text{O}=\text{O}) = 614 + 4(\text{C}-\text{H}) + 1485$ (1) bonds made $4(\text{C}=\text{O}) + 4(\text{O}-\text{H}) = 3196 + 1860 = 5056$ (1) $2099 + 4(\text{C}-\text{H}) - 5056 = -1387$ (1) average C—H bond enthalpy = $\frac{1570}{4} = 392.5 / 393 \text{ kJ mol}^{-1}$ (1)		4		4	3	
		(ii)		ethene is a gas / not a liquid			1	1		1
				Question 7 total	2	13	3	18	7	8